Chapter 29 Review

Campbell Biology, AP Edition, 12e

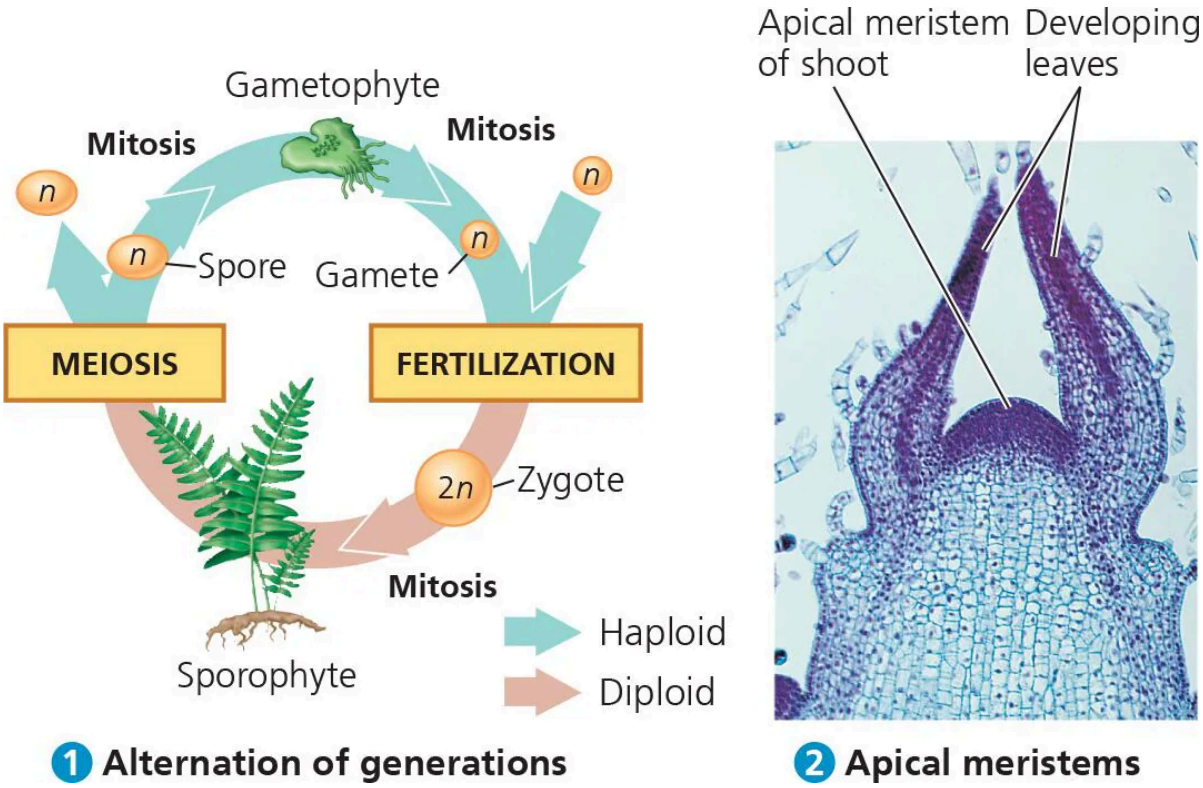
Chapter 29 Review

Summary of Key Concepts

As plants began to colonize land, how was the cycling of matter on Earth affected? **(BIG IDEAS 2 & 4)**

Concept 29.1: Plants evolved from green algae

- Morphological and biochemical traits, as well as similarities in nuclear and chloroplast genes, indicate that certain charophyte algae are the closest living relatives of plants.
- A protective layer of ****__sporopollenin__ and other traits allow charophytes to tolerate occasional drying along the edges of ponds and lakes. Such traits may have enabled the algal ancestors of plants to survive in terrestrial conditions, opening the way to the colonization of dry land.
- Derived traits that distinguish plants from charophytes, their closest algal relatives, include ****__cuticles__, ****__stomata__, multicellular dependent embryos, walled spores produced in sporangia, and the two shown here:



1. Alternation of generations diagram in which the gametophyte (n) produces haploid gametes by mitosis. Gamete (n) from another plant is introduced and the two gametes unite (fertilization) and form a diploid zygote. The zygote (2 n) develops into a multicellular diploid sporophyte by mitosis. The sporophyte produces unicellular haploid spores by meiosis. The spores (n) develop into multicellular haploid gametophytes and the cycle begins again.
2. Apical meristems showed with the apical meristem of the shoot and developing leaves.

- Fossils show that plants arose more than 470 million years ago. Subsequently, plants diverged into several major groups, including nonvascular plants (****__bryophytes__); ****__seedless vascular__

plants, such as ****lycophytes** and ferns; and the two groups of seed plants: ****gymnosperms** and ****angiosperms**.

Draw a phylogenetic tree illustrating our current understanding of plant phylogeny; label the common ancestor of plants and the origins of vascular tissue, apical meristems, and seeds.

Concept 29.2: Mosses and other nonvascular plants have life cycles dominated by gametophytes

- Lineages leading to the three extant clades of nonvascular plants, or ****bryophytes*—***liverworts**, ****mosses**, and ****hornworts**—diverged from other plants early in plant evolution.
- In bryophytes, the dominant generation consists of haploid ********gametophytes**, such as those that make up a carpet of moss. ****Rhizoids** anchor gametophytes to the substrate on which they grow. The flagellated sperm produced by ****antheridia** require a film of water to travel to the eggs in the ****archegonia**.
- The diploid stage of the life cycle—the ********sporophytes**—grow out of archegonia and are attached to the gametophytes and dependent on them for nourishment. Smaller and simpler than vascular plant sporophytes, they typically consist of a ****foot**, ****seta** (stalk), and ********sporangium**.
- *Sphagnum*, or peat moss, is common in large regions known as peatlands and has many practical uses, including as a fuel.

Summarize the ecological importance of mosses.

Concept 29.3: Ferns and other seedless vascular plants were the first plants to grow tall

- Fossils of the forerunners of today's vascular plants date back about 425 million years and show that these small plants had independent, branching sporophytes and a vascular system.
- Over time, other derived traits of living vascular plants arose, such as a life cycle with dominant sporophytes, lignified vascular tissue, well-developed ****roots** and **leaves**, and ****sporophylls**.
- Seedless vascular plants include the ****lycophytes** (phylum Lycophyta: club mosses, spikemosses, and quillworts) and the ****monilophytes** (phylum Monilophyta: ferns, horsetails, and whisk ferns and relatives). Current evidence indicates that seedless vascular plants, like bryophytes, do not form a clade.
- Ancient lineages of lycophytes included both small herbaceous plants and large trees. Present-day lycophytes are small herbaceous plants.

- Seedless vascular plants formed the earliest forests 385 million years ago. Their growth may have contributed to a major global cooling that took place during the Carboniferous period. The decaying remnants of the first forests eventually became coal.

What trait(s) allowed vascular plants to grow tall, and why might increased height have been advantageous?